

DOC

[Doctoral Fellowship Program of the Austrian Academy of Sciences]

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Review

Name of the Applicant

Zishu Luo

Please assess the following aspects of the research project by allocating marks on a scale of 1 to 10 and explain in your written statement for each question the reasons for your evaluation.

Comment on the rating scale

9-10 = outstanding

7-8 = very good

5-6 = good

3-4 = average

1-2 = poor

1. Evaluation of the research project

a) How do you assess the originality, the degree of innovation and relevance of the research project with regard to the international research in the field? Please elaborate.

Score: 8 / 10

The proposal is timely in that it combines three strands: BEM development, surrogate modeling, and an XR-facing system. These are combined into an end-to-end pipeline. The core innovative angle is less about new techniques within any single module, and more about integrating them to enable real-time analysis.

That said, the project seems to aim for breadth over depth, which is the main weakness. Each module is substantial and carries certain risks; bundling all three into one funded project increases the probability that at least one component under-delivers, which can then compromise the entire end-to-end vision.

b) In your opinion, has the applicant been able to sum up the current state of research in their discipline including recent developments? Please elaborate.

Score: 7 / 10

The applicant's related work is extensive and demonstrates strong engagement with the domain literature. However, the framing appears to draw primarily from domain venues rather than integrating the most relevant computer science / ML / XR perspectives where appropriate. This matters because the project explicitly spans these areas. For an XR perspective, it would be interesting to cover state-of-the-art interaction paradigms, evaluation methodologies.

Relevant XR works include:

- 1) DXR on a toolkit to facilitate the creation of visualizations in XR:
<https://ieeexplore.ieee.org/abstract/document/8440858>
- 2) A framework for evaluation of immersive systems:
<https://link.springer.com/article/10.1007/s10055-021-00618-y>

In short, the applicant demonstrates strong coverage of the domain state of the art, but the proposal would be strengthened by a more balanced and up-to-date synthesis across the computational + ML + XR strands that the project claims as core.

c) Are the research questions clearly defined? Are the theoretical assumptions and the methods that have been chosen to answer the research questions the best way to accomplish this task? Are they described in sufficient detail? Please elaborate.

Score: 10 / 10

The research questions are generally understandable and appear aligned with the proposed modules.

d) Are the individual parts of the project planned in a meaningful and sensible way? Is it possible to carry out the project within the time schedule? Please elaborate.

Score: 8 / 10

The plan is well organized and sensibly chunked into annual phases, which is a strong point for feasibility and accountability. The sequencing also appears reasonable: early foundational work, followed by model development, followed by integration and demonstration.

The main improvement would be to include a mitigation/contingency plan, since the end-to-end success depends on multiple high-risk components. A simple risk assessment would already help: identify top risks, early warning indicators, and fallback options.

2. Final statement and overall evaluation with a recommendation for or against funding the application:

Overall, I view this as a strong, internationally relevant proposal with high potential impact due to its end-to-end vision and interdisciplinary integration. The applicant demonstrates clear competence and a reasonable execution plan. The primary concern is scope risk: three major modules, each difficult, could diffuse contributions and increase the chance that one weak link limits the overall outcome.

Total Score: 33 / 40

Recommend funding with highest priority	☐
Recommend funding if sufficient funds are available	☒
Recommend refusal of the project in its current version	☐

Your written statement will be handed out in anonymous form to the applicant. Please tell us if you do not approve of this procedure.

I am aware that the project proposal and all supporting materials are confidential; I deleted the proposal after reviewing it. I observed the requirement of reviewer anonymity during the entire review process.

Legal basis for transmission of data for your reviewer assignment is the Research Organisation Act (FOG). Since in your work as a reviewer you will be processing, inter alia, confidential personal data, please find a summary of the relevant regulations concerning data protection for reviewers on our [website](#).